Section	Title	Description	length	Example write-up
General Information	System Name			
	Versioning Information	Date of system’s first release Latest version Latest release date Include details of how versions are being tracked and updated		
	Primary Developer/Org	Individual/Org developing the system		
	Contact Info	Email/phone number/address of the primary Developer/Org		
	System Overview	Short description of uses and the system itself(e.g., AI models, datasets, APIs) and their roles)		AI-powered customer tracking system that uses computer vision models to analyze video streams from security cameras for customer traffic, dwell time, and queue analytics without facial recognition. It processes data through on-premise edge computing and cloud-based APIs (Azure AI Vision), providing real-time heatmaps and automated alerts via a retail analytics dashboard.
Intended Purpose and Scope	Primary intended uses			• Customer Flow Optimization: Identifying high-traffic zones to improve store layouts. • Queue Management: Monitoring waiting times at checkout counters to optimize staffing. • Dwell Time Analysis: Measuring how long customers spend in different sections of the store.
	Primary intended users			• Retail Store Managers & Owners: Optimizing store layout and staffing based on foot traffic data. • Marketing & Operations Teams: Understanding customer movement for better promotions and engagement. • Security Teams: Monitoring occupancy levels in compliance with safety regulations.
	Out-of-scope use	Use cases where		• Facial Recognition or Identifiable

	cases	system is not applicable		Tracking: System is not designed to track or identify specific individuals. ● Employee Productivity Monitoring: Not intended for surveillance of staff performance. ● Law Enforcement & Surveillance: Cannot be used for forensic tracking or law enforcement without explicit regulatory approval.
Existing doc	Terms and conditions	Existing terms and conditions document: e.g. Usage Terms, Limitations of Liability, License Type, Third-party, Dependencies, Privacy and Data Handling, Compliance Acknowledgment		
	Current legal compliance status	Existing legal assessment result for policy compliance(if any), e.g., risk level		● GDPR (EU General Data Protection Regulation): Compliant with privacy-by-design principles, ensuring no personally identifiable information (PII) is stored. ● ISO/IEC 27701 (Privacy Information Management): Meets privacy management system standards.
Training Data Information	Dataset Description	Summarization of dataset used for training/validation, e.g. purpose/size/key/ characteristics		Comprises 1.2 million hours of video footage from publicly available surveillance datasets, synthetic video augmentation, and licensed proprietary sources.
	Collection Method	Details how the dataset was collected, including the sources, consent processes and methodologies (e.g., scraping, surveys)		The primary sources include: ● DIVA Dataset (DARPA Video Analysis Benchmark) – Provides annotated surveillance footage for activity recognition tasks. ● AI City Challenge Dataset (NVIDIA) – Contains retail and urban surveillance videos for pedestrian tracking. ● Synthetic Data (Computer-Generated Simulations) – Used to generate edge cases (e.g., varied lighting, occlusions).

				Licensed Data from Partner Retailers – Video collected with explicit business agreements ensuring customer privacy protections. The dataset does not store personally identifiable information (PII) and follows privacy-preserving techniques such as blurring and anonymization.
	Bias Mitigation Measures	Steps taken to identify and address potential biases in the dataset, e.g. over/underrep		- Balanced dataset collection from stores in different socioeconomic regions. - Model testing across diverse lighting conditions, store layouts, and cultural contexts to prevent bias in movement detection. - Hired third-party audits to assess potential biases in crowd flow predictions.
	Usage Constraints	Short description of licensing restrictions/privacy concerns/appropriate application domains		- Cannot be used for individual tracking or face recognition. - Retailers must disclose AI usage to customers via in-store notices.
Model Performance and Evaluation	Summary of Performance Assessment	Quantitative measures to evaluate how well the model performs its intended task, e.g. accuracy/precision/recall/fairness scores.		- Customer Movement Detection Accuracy: 94.2% - Queue Length Estimation Error: +-2 people per queue - Dwell Time Measurement Accuracy: +-5% compared to human-annotated benchmarks
	Disaggregated Performance（Mitchell et al）	Performance results broken down by demographic subgroups or other relevant categories to identify biases or disparities.		- Average Adult (5'4" – 6'2") 94.5% - Shorter Adults (<5'4") 91.2% - Tall Adults (>6'2") 92.8% - Children (<4'5") 87.3% - People in Wheelchairs 85.6% - Crowded Environments 88.9% - Obstructed Views 86.4%
	Testing Contexts	Describes the scenarios, environments, or specific use cases where the model was tested, ensuring relevance to real-world applications		- Evaluated across 50+ retail locations to test model consistency. - Tested for peak and off-peak hours to measure real-time performance fluctuations. - Validated for indoor, semi-outdoor, and open-space environments.

	Evaluations for Edge Cases or Adversarial Inputs	Testing the model on rare or challenging scenarios, including adversarial inputs designed to probe weaknesses.		• Performance under crowd surges: Stress-tested with synthetic and real event footage. • Occlusion handling: Tested for movement detection in cases where multiple customers overlap in view. • Adversarial spoofing (fake humans, mannequins): Model showed resilience against false positives.
Ethical Consideratio ns	Potential Risks and Harms	Any risks or harms that could arise from the model's use, e.g. unintended consequences/ failures in high-stakes applications.		• Risk of Privacy Violations: Potential for system misuse in surveillance applications. • Crowd Prediction Errors: Incorrect analysis could result in misallocation of store resources. • Algorithmic Bias: Risk of skewed customer behavior analysis due to underrepresented store types in training data.
	Actions taken	Outline any steps taken to address the above issues.		• Ensuring accessibility for disabled individuals: Training data includes footage of wheelchair users to ensure fair detection. • Minimizing retail discrimination risks: The system does not infer customer demographics, age, or gender. • Regulatory Review Mechanisms: Quarterly audits ensure no unintended discriminatory outcomes.
	Misuse Scenarios	Describe scenarios where the model could be used but misused, either intentionally or unintentionally, and explain actions taken to minimize risks		• Unauthorized Employee Tracking: Retailers cannot use the system for employee surveillance. • Use in Public Spaces Without Consent • Integration with Facial Recognition: The system does not support or process identifiable personal data. • Mitigation: Implemented strict access controls and compliance audits.
Maintenance and Monitoring	Human Oversight	Details the role of humans in monitoring and intervening		• Store managers have real-time dashboards to monitor predictions and override incorrect detections. • Anomaly detection alerts notify human operators of unusual behavior patterns. • Independent human reviews ensure that crowd flow assessments remain unbiased.

	Update Frequency	Description of how often the system or model is updated(Post-deployment Monitoring)		• Model retraining every 6 months to incorporate new data from partner retailers. • Compliance reviews every 12 months to ensure alignment with evolving privacy regulations. • Real-time bug monitoring system for immediate performance issue detection.